NEUROX 3D NEUROLOGICAL BRAIN SEGMENTATION

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ABSTRACT:

NeuroX is a unified artificial intelligence framework for multi-disease neurological MRI analysis designed to detect brain tumors, stroke lesions, and Alzheimer's disease within a single inference pipeline. Existing clinical AI systems typically focus on one neurological disorder at a time, creating fragmented diagnostic workflows and limiting clinical efficiency. NeuroX addresses this limitation through a hybrid architecture combining 3D convolutional neural networks, transformer bottleneck learning, attention-guided lesion segmentation, uncertainty-aware prediction, and automated report generation. The framework processes volumetric MRI data after normalization, skull stripping, and spatial resampling to standardized dimensions. Tumor and stroke tasks share a common encoder-decoder path, while Alzheimer classification uses an independent residual branch optimized for diffuse structural pattern recognition. Experimental evaluation on BraTS, ISLES, and ADNI datasets demonstrates strong segmentation and classification performance, with tumor whole-tumor Dice reaching 0.8902 and stable multi-task learning under curriculum training. NeuroX further generates 3D lesion visualizations and automated diagnostic summaries, improving interpretability and deployment readiness.

Keywords: Brain MRI, 3D CNN, Transformer, Tumor Segmentation, Stroke Detection, Alzheimer's Classification, Medical A

Objective:

The main objective of this research is to develop NeuroX, a unified artificial intelligence framework for comprehensive brain MRI analysis capable of detecting multiple neurological disorders within a single diagnostic pipeline. The study aims to overcome the limitations of existing single-task medical imaging systems by integrating brain tumor detection, stroke lesion identification, and Alzheimer's disease classification into one robust architecture. To achieve high diagnostic precision, the proposed framework employs advanced MRI preprocessing techniques, including normalization, skull stripping, and spatial resampling, followed by deep volumetric feature extraction using a hybrid 3D Convolutional Neural Network (3D CNN) and Transformer-based architecture. Another important objective is to perform accurate lesion segmentation for tumor and stroke regions using an attention-guided decoder while simultaneously analyzing diffuse neurodegenerative patterns associated with Alzheimer's disease through a dedicated residual learning branch. In addition, the research focuses on improving clinical reliability by incorporating uncertainty estimation techniques, allowing the model to provide confidence-aware predictions for medical decision support. The framework also aims to generate 3D lesion visualization and automated clinical reports, making the system more practical for real-world deployment in radiology workflows and AI-assisted neurological diagnosis.

Methodology:

The proposed NeuroX framework follows a structured multi-stage methodology designed to process brain MRI data and perform unified neurological disease analysis within a single intelligent pipeline. The process begins with MRI data acquisition from multiple benchmark datasets corresponding to brain tumor, stroke, and Alzheimer's disease cases. Since MRI scans obtained from different sources often vary in intensity distribution, spatial resolution, and modality characteristics, an initial preprocessing stage is applied to standardize the data before model training. This preprocessing includes voxel intensity normalization using Z-score normalization to reduce scanner-based variation, followed by skull stripping to remove non-brain tissues and retain only the relevant anatomical region. After preprocessing, all MRI volumes are resampled and padded to a fixed spatial size of 96 × 96 × 96 so that each input volume maintains dimensional consistency across tasks.

Once standardized, the MRI data are passed into the feature extraction stage, where a three-dimensional convolutional neural network is used to capture volumetric spatial information from the brain scans. Unlike traditional two-dimensional convolution methods, the 3D convolutional layers preserve inter-slice anatomical relationships, which are essential for detecting neurological abnormalities. The extracted deep features are then forwarded into a transformer bottleneck module that models long-range spatial dependencies across the full brain volume. This combination of local convolutional learning and global transformer attention enables the system to capture both fine lesion details and broader structural context.

For disease-specific analysis, the architecture follows a dual-path strategy. Brain tumor and stroke tasks share a common encoder backbone and are processed through attention-guided decoder layers that generate segmentation masks for abnormal regions. These decoder layers selectively focus on medically relevant feature maps and improve boundary localization for lesions. In parallel, Alzheimer's disease analysis is performed through a dedicated residual branch optimized for identifying diffuse structural degeneration patterns rather than localized lesions.

The framework also incorporates uncertainty estimation to improve prediction reliability. During inference, Monte Carlo dropout is applied across multiple forward passes to estimate confidence variation, allowing the model to identify cases where predictions are uncertain and may require expert review. After segmentation, lesion masks are converted into three-dimensional surface structures using geometric reconstruction techniques, enabling clear anatomical visualization. Finally, the extracted predictions, lesion measurements, and confidence scores are combined into an automated diagnostic report, making the system suitable for clinical decision-support applications. This methodology enables NeuroX to function as an integrated, scalable, and clinically interpretable neurological MRI analysis system.

Implementation:

The implementation of NeuroX was carried out using a modular deep learning pipeline developed in Python, where each stage of the system was designed as an independent yet interconnected module for efficient neurological MRI analysis. The project begins with MRI data loading using NiBabel, which enables handling of volumetric NIfTI files commonly used in medical imaging datasets. After loading, preprocessing operations such as intensity normalization, resizing, padding, and skull stripping are applied to prepare the MRI scans for model input. Skull stripping was integrated using HD-BET to remove non-brain tissues and improve feature consistency. All MRI volumes were then resized to a fixed dimension of 96 × 96 × 96 to ensure compatibility during training and inference.
The core model was implemented using PyTorch, where a hybrid architecture combining three-dimensional convolutional layers and transformer-based learning was developed. The shared encoder extracts volumetric spatial features for tumor and stroke tasks, while a transformer bottleneck captures long-range anatomical dependencies across the brain volume. For lesion segmentation, an attention-guided decoder was implemented to generate precise masks for tumor and stroke regions. A separate residual branch with squeeze-and-excitation blocks was designed specifically for Alzheimer's disease classification, allowing the model to focus on global structural degeneration patterns. To improve reliability during inference, Monte Carlo dropout layers were activated during prediction to estimate uncertainty and confidence levels for each diagnostic output.
Training was performed on a NVIDIA Tesla T4 GPU using a phased curriculum strategy over 48 epochs, where disease-specific branches were gradually optimized before joint training. Due to the large memory requirement of 3D MRI processing, batch size was limited to one with gradient accumulation to stabilize updates. The loss function combined Dice loss and binary cross-entropy to balance segmentation accuracy and classification learning. For visualization, segmented lesions were converted into 3D surface structures using marching cubes and displayed interactively with Plotly. Finally, the complete system was deployed through Streamlit, enabling MRI upload, disease prediction, lesion visualization, and automated PDF clinical report generation within a user-friendly interface.

I. INTRODUCTION

1.1. Background and Motivation

The rapid advancement of artificial intelligence in medical imaging has significantly transformed the way neurological disorders are detected and analyzed. Brain diseases such as tumors, ischemic stroke, and Alzheimer's disease require early diagnosis because delayed interpretation may directly affect treatment success and patient survival. Magnetic Resonance Imaging (MRI) is widely used for neurological diagnosis because it provides high-resolution structural information of brain tissues. However, interpreting volumetric MRI scans manually remains time-consuming and requires highly specialized radiological expertise.

Traditional computer-aided diagnostic systems generally focus on one disease at a time, which forces clinicians to rely on multiple independent tools for complete neurological assessment. This fragmented workflow reduces efficiency and increases interpretation complexity. In addition, many existing systems provide only classification output without lesion localization or confidence estimation, limiting practical adoption in clinical environments.

To address these limitations, NeuroX was developed as a unified framework capable of analyzing multiple neurological conditions within one diagnostic pipeline. The project integrates tumor detection, stroke segmentation, Alzheimer's classification, uncertainty estimation, 3D visualization, and automated report generation in a single deployable architecture.

1.2. Problem Statement

Most existing medical imaging AI systems are designed as isolated disease-specific solutions. Tumor segmentation systems cannot detect neurodegenerative disorders, while Alzheimer classification systems cannot identify acute lesions such as stroke. Separate models create inconsistent outputs and require multiple inference pipelines.

Another major limitation is the absence of uncertainty awareness. In clinical applications, confidence estimation is essential because incorrect predictions without reliability indicators can lead to unsafe decision-making. NeuroX addresses these limitations by combining shared deep feature extraction with task-specific disease branches.

1.3. Project Contributions

This research provides several contributions:

- Development of a unified multi-disease MRI analysis framework
- Integration of 3D CNN and Transformer learning
- Attention-guided lesion segmentation for tumor and stroke
- Confidence-aware predictions using Monte Carlo Dropout
- Automated clinical report generation
- Interactive 3D lesion visualization

1.4. Paper Structure

- Section II presents literature survey
- Section III explains methodology and architecture
- Section IV describes implementation
- Section V discusses results
- Section VI concludes the study

II. LITERATURE SURVEY/ RELATED WORKS

A. Evolution of Medical Imaging AI

The development of artificial intelligence in medical imaging has progressed through multiple stages, beginning with traditional image processing techniques and gradually evolving toward deep learning-based automated diagnosis systems. In the early stages, medical image analysis mainly depended on handcrafted feature extraction methods, where domain experts manually designed texture descriptors, intensity-based measures, morphological features, and statistical attributes to represent abnormalities in medical images. These handcrafted features were then supplied to classical machine learning algorithms such as support vector machines, decision trees, and random forests for classification tasks. Although these approaches produced acceptable performance for limited datasets, their effectiveness depended heavily on the quality of manually selected features and expert knowledge. In complex neurological MRI analysis, handcrafted approaches often failed to capture subtle structural variations because brain abnormalities can appear with highly variable shapes, sizes, and intensity distributions across patients. In addition, traditional methods struggled to generalize when applied to MRI data collected from different scanners, hospitals, or imaging protocols. As medical imaging datasets increased in size and computational resources improved, researchers gradually shifted toward representation learning methods where neural networks automatically learned relevant diagnostic features directly from raw images. This transition marked a major breakthrough because deep learning models removed the need for manual feature engineering and demonstrated improved robustness across diverse medical imaging tasks.

B. B. 3D CNN in MRI Analysis

Three-dimensional convolutional neural networks introduced a major advancement in brain MRI analysis because they process volumetric medical data directly rather than treating each MRI slice independently. Unlike two-dimensional convolutional networks, which analyze individual image slices separately, 3D CNN architectures preserve spatial continuity across adjacent slices and therefore capture anatomical relationships throughout the full brain volume. This property is especially important for neurological disorders because lesions, tumors, and tissue degeneration often extend across multiple slices and require volumetric interpretation. In brain tumor research, 3D CNN models demonstrated strong segmentation capability by learning local texture variations and lesion boundaries directly from MRI volumes. Their ability to capture volumetric context significantly improved segmentation accuracy for tumor subregions such as enhancing tumor, tumor core, and whole tumor regions. Similarly, 3D CNNs have been applied to stroke lesion detection, where they help identify ischemic regions that may occupy irregular three-dimensional patterns within the brain. In Alzheimer's disease analysis, 3D CNNs became useful for identifying subtle structural degeneration patterns across cortical regions and hippocampal areas. Despite these strengths, pure 3D CNN models still have limitations because convolution kernels mainly capture local neighborhoods and may fail to model long-range anatomical dependencies across distant brain regions. This limitation becomes important when diseases involve distributed structural changes rather than highly localized lesions. Furthermore, deeper 3D CNN architectures require significant GPU memory because volumetric MRI processing is computationally expensive.

C. Transformer Integration

Transformers were introduced into medical imaging to overcome the limitations of convolutional models in capturing long-range dependencies. Originally developed for natural language processing, transformer architectures use self-attention mechanisms that allow the model to analyze relationships between distant regions within input data. When applied to MRI analysis, transformers enable global contextual learning by identifying interactions between anatomically distant structures that may jointly contribute to disease patterns. In neurological imaging, this global attention mechanism becomes particularly valuable because certain disorders involve distributed structural changes across multiple brain regions rather than isolated lesions. For example, Alzheimer's disease often produces gradual cortical thinning and hippocampal degeneration distributed across multiple anatomical regions, which benefits from transformer-based context learning. Similarly, tumor infiltration patterns and stroke-related tissue changes can extend beyond local lesion boundaries, requiring broader contextual interpretation. Hybrid CNN-transformer systems emerged as an effective solution because convolutional layers first extract stable local spatial features, while transformer bottlenecks refine these representations by modeling long-range relationships. Although transformer integration significantly improves feature richness, transformer-only architectures remain computationally demanding because self-attention operations scale heavily with input volume size. For full 3D MRI data, this creates high memory requirements and slower training speed, limiting real-time deployment unless carefully optimized.

D. Attention U-Net

Attention U-Net represents an important advancement in medical image segmentation because it extends the conventional U-Net architecture by introducing attention gates that selectively emphasize clinically important features during decoding. In standard U-Net models, skip connections transfer encoder features directly to decoder layers, but not all transferred information contributes equally to lesion segmentation. Attention gates solve this limitation by learning to suppress irrelevant background features while highlighting spatial regions likely to contain abnormalities. This mechanism improves segmentation accuracy, especially in cases where lesions occupy small regions relative to the full brain volume. In tumor segmentation, attention-guided decoding improves boundary precision for enhancing tumor regions and necrotic cores. For stroke lesion analysis, attention mechanisms help isolate small ischemic areas that may otherwise be overlooked due to severe class imbalance between lesion and healthy tissue. Attention U-Net models also improve robustness when lesion shapes are irregular or when intensity contrast is weak. Because medical MRI data often contain complex tissue patterns, attention-based feature refinement provides better localization than conventional encoder-decoder architectures. However, most existing attention-based systems remain disease-specific and usually focus on only one segmentation task at a time, limiting broader clinical application.

E. Research Gap

Despite major progress in medical imaging AI, several important gaps remain unresolved in current neurological diagnostic systems. Most published models focus on only one disease category, such as brain tumor segmentation, stroke lesion identification, or Alzheimer's classification, and therefore cannot provide comprehensive neurological assessment within one inference framework. In practical clinical settings, however, multiple neurological abnormalities may coexist or require simultaneous screening during MRI evaluation. Existing systems also commonly separate classification and segmentation tasks into independent pipelines, increasing computational redundancy and reducing deployment efficiency. Another major limitation is the lack of uncertainty estimation in most medical AI models. Clinical decision support systems require confidence-aware outputs because incorrect predictions without reliability indicators may create serious diagnostic risks. Furthermore, many current research models focus mainly on academic performance metrics while lacking practical deployment features such as 3D visualization and automated report generation. NeuroX addresses these gaps by proposing a unified architecture that combines multi-disease detection, lesion segmentation, uncertainty estimation, volumetric visualization, and automated reporting within a single clinically oriented framework.

III. METHODOLOGY AND SYSTEM ARCHITECTURE

A. System Workflow

The methodology of NeuroX was designed as a complete end-to-end diagnostic pipeline capable of transforming raw volumetric MRI data into clinically interpretable neurological predictions, lesion localization outputs, confidence measurements, and automated diagnostic summaries. Unlike conventional medical imaging systems that separate preprocessing, classification, segmentation, and reporting into independent components, NeuroX integrates every stage within one structured framework so that all disease-specific outputs emerge from a unified computational process.

The workflow begins with MRI acquisition, where volumetric scans from different neurological datasets are collected in NIfTI format because this format preserves voxel-level spatial information together with anatomical affine orientation required for later reconstruction. Since tumor, stroke, and Alzheimer datasets originate from different institutions and imaging protocols, each dataset initially presents variation in intensity distribution, voxel spacing, modality composition, and anatomical orientation. Without standardization, these differences can introduce severe training instability and prevent feature consistency across tasks.

After MRI acquisition, the preprocessing stage converts heterogeneous raw scans into a uniform representation suitable for deep learning. This stage is extremely important because medical imaging models are highly sensitive to inconsistencies in scanner acquisition and voxel statistics. Once preprocessing is completed, the normalized volumes are passed into the feature extraction stage, where the deep learning architecture begins hierarchical representation learning.

The extracted features are then routed into disease-specific diagnostic paths. NeuroX uses a shared multi-task learning design in which tumor and stroke tasks share one volumetric feature backbone because both rely heavily on focal lesion understanding, while Alzheimer's disease follows a dedicated branch because it depends more on diffuse anatomical degeneration rather than localized lesions.

After disease-specific feature routing, the segmentation stage produces lesion masks for tumor and stroke abnormalities. Segmentation is required because clinical interpretation needs precise lesion boundaries rather than only disease labels. These masks are then used for volumetric analysis and anatomical measurement.

The uncertainty estimation stage follows prediction generation. This component determines how reliable each model output is by repeatedly sampling the prediction space during inference. Confidence estimation is clinically important because uncertain cases must be flagged for expert review.

Following uncertainty analysis, segmented lesion outputs are converted into visual representations through geometric reconstruction methods that generate three-dimensional lesion surfaces. Finally, all outputs including disease probability, lesion volume, uncertainty score, and anatomical interpretation are assembled into an automated clinical report designed to support radiological decision-making.

Thus, the full NeuroX workflow consists of eight tightly integrated stages:

- MRI acquisition
- Standardized preprocessing
- Volumetric feature extraction
- Multi-disease prediction
- Lesion segmentation
- Uncertainty estimation
- 3D visualization
- Automated report generation

This workflow ensures that raw imaging data can be transformed into clinically meaningful neurological intelligence within one unified inference pipeline.

B. Dataset Description

The performance and generalization capability of NeuroX depend strongly on the diversity and quality of training data. To support multi-disease learning, three internationally recognized benchmark datasets were selected, each corresponding to one neurological condition. These datasets were chosen because they are widely accepted in medical AI research and provide validated MRI annotations suitable for supervised learning.

Brain Tumor Dataset: BraTS 2020 + 2021

For tumor analysis, NeuroX uses the Medical Image Computing and Computer Assisted Intervention Society BraTS 2020 and 2021 datasets, which represent one of the most widely used international benchmarks for glioma segmentation.

The BraTS dataset contains multi-modal MRI scans including:

- T1
- T1ce
- T2
- FLAIR

Each modality provides complementary anatomical information:

- T1 shows structural anatomy
- T1ce highlights enhancing tumor tissue
- T2 captures edema
- FLAIR suppresses fluid and improves lesion visibility

Each tumor case includes expert voxel-wise annotations for:

- Enhancing tumor (ET)
- Tumor core (TC)
- Whole tumor (WT)

A total of 1619 scans were used across both challenge versions, enabling strong representation of tumor heterogeneity.

Stroke Dataset: ISLES 2022

Stroke lesion detection uses ISLES Challenge ISLES 2022, which focuses on ischemic stroke lesion segmentation.

Stroke MRI differs significantly from tumor MRI because lesions are often small, irregular, and highly variable in intensity depending on infarction stage.

The dataset includes:

- Diffusion-weighted imaging
- Perfusion-related modalities
- Lesion masks

Stroke data introduces strong class imbalance because lesion voxels occupy very small fractions of the total brain volume.

A total of 250 stroke scans were used.

Alzheimer Dataset: ADNI

For Alzheimer classification, NeuroX uses Alzheimer's Disease Neuroimaging Initiative ADNI, one of the most important longitudinal neurodegenerative imaging repositories.

ADNI focuses on structural MRI for neurodegenerative disease progression and includes:

- Healthy controls
- Mild cognitive impairment
- Alzheimer’s disease cases

A total of 1208 MRI volumes were used.

Unlike tumor and stroke datasets, ADNI does not focus on lesion segmentation but on distributed structural degeneration patterns.

C. Preprocessing Pipeline

MRI preprocessing is one of the most critical stages because raw scans from different datasets cannot be directly supplied into a unified model.

1. **Z-score Normalization**

MRI intensities are not standardized across scanners. The same tissue may appear with different voxel intensity values depending on acquisition hardware and protocol.

Therefore, voxel intensities are normalized using:

$$Z = \frac{(X - \mu)}{\sigma}$$

Where:

- X = voxel intensity
- μ = mean intensity
- σ = standard deviation

This reduces scanner variability.

2. **Skull Stripping**

Brain extraction is required because non-brain tissues introduce irrelevant information.

NeuroX applies HD-BET to remove:

- Skull
- Skin
- Fat tissue
- Background

This improves anatomical focus.

3. Spatial Resampling

MRI volumes differ in size across datasets.

All scans are resampled into:

- 96 × 96 × 96 voxels

This standardization reduces memory instability and enables batch consistency.

4. Channel Alignment

Tumor and stroke use multi-channel input.

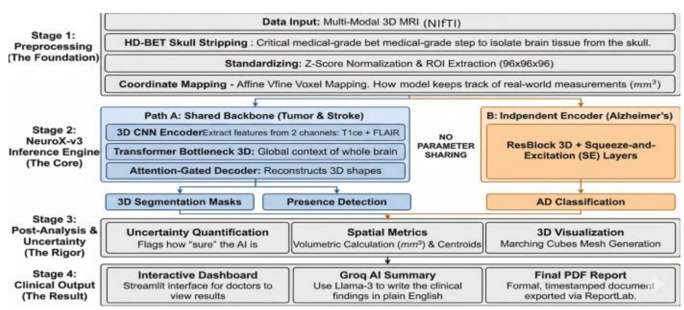
Example:

- T1ce
- FLAIR

These are stacked as multi-channel volumetric tensors.

Alzheimer uses single-channel anatomical MRI because diffuse degeneration relies on structural intensity distribution.

D. Core Learning Architecture



NeuroX uses a hybrid architecture combining local volumetric learning and global contextual modeling.

- Shared Clinical Backbone

Tumor and stroke tasks share a common volumetric encoder because both require focal lesion analysis.

- 3D CNN Encoder

The encoder uses stacked 3D convolutions to learn:

- Local textures
- Spatial lesion boundaries
- Tissue intensity gradients

3D convolution preserves volumetric continuity.

Each encoder block contains:

- 3D convolution

- Instance normalization
- ReLU activation

Pooling reduces spatial dimensions progressively.

- Transformer Bottleneck

After deep convolutional encoding, features enter a transformer bottleneck.
The transformer captures:

- Long-range anatomical dependencies
- Inter-regional contextual relationships

This solves the limitation of local-only convolution learning.
Self-attention improves global interpretation.

- Segmentation Path

Tumor and stroke segmentation use an attention-guided decoder.

- Attention-Gated U-Net Decoder

Decoder stages include:

- Upsampling
- Skip connections
- Attention gates

Attention gates suppress irrelevant tissue and emphasize lesion regions.
This improves:

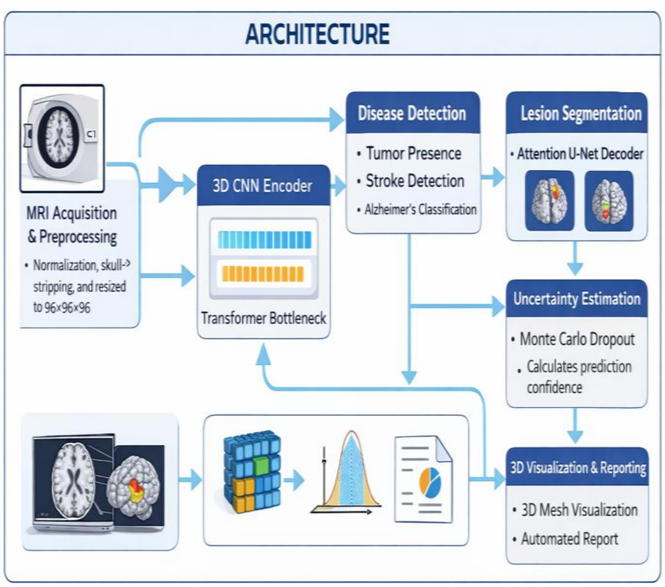
- Small lesion detection
- Boundary sharpness
- Noise suppression
- Alzheimer Branch

Alzheimer detection follows an independent branch because neurodegeneration is diffuse rather than focal.

- Residual CNN with SE Blocks

Residual learning stabilizes deeper layers.
Squeeze-and-excitation blocks improve channel importance weighting.
This branch focuses on:

- Cortical thinning
- Ventricular expansion
- Hippocampal degeneration



E. Uncertainty Estimation

Clinical AI must provide reliability, not only prediction.
NeuroX uses Monte Carlo Dropout during inference.
Instead of one deterministic forward pass, the model performs repeated stochastic passes.
Each pass produces slightly different outputs.
Prediction variance estimates uncertainty.
High variance indicates uncertain diagnosis.
Low variance indicates stable confidence.
This separates:

- Epistemic uncertainty (model uncertainty)
- Aleatoric uncertainty (data uncertainty)

Confidence estimation is especially important in:

- borderline tumors
- very small stroke lesions
- early Alzheimer structural changes

Cases with high uncertainty are flagged for clinical review.

IV. IMPLEMENTATION

A.Model Training and Evaluation Pipeline

Purpose

The model training and evaluation pipeline of NeuroX serves as the central computational engine responsible for learning disease-specific representations from volumetric MRI data, validating prediction performance, and preparing trained components for deployment inside the final clinical inference system. Unlike conventional machine learning pipelines where training and deployment are often separated by multiple manual stages, NeuroX follows an automated structured training workflow in which preprocessing, model optimization, validation, and checkpoint generation are integrated into one controlled experimental environment. The training pipeline was designed as an offline computational process because three-dimensional MRI learning requires substantial GPU resources, repeated iterative optimization, and large-scale tensor operations that are not suitable for direct real-time execution inside a web interface.
The training system begins by loading volumetric MRI datasets corresponding to tumor, stroke, and Alzheimer's disease into memory using NiBabel, which allows voxel-level extraction from NIfTI files while preserving affine metadata required for later reconstruction. After MRI loading, all volumes pass through the preprocessing stages defined earlier, including normalization, skull stripping, spatial resizing, and channel alignment. This preprocessing is executed before every training epoch so that the model always receives standardized input regardless of original scanner characteristics.
Once preprocessing is completed, the model architecture is initialized using **PyTorch**. The shared encoder, transformer bottleneck, segmentation decoder, and Alzheimer branch are instantiated together as a unified multi-task network. During each training iteration, volumetric input tensors are passed through the encoder to extract hierarchical features,

followed by transformer attention refinement and disease-specific branch routing.

The optimization process uses a curriculum-based multi-phase strategy. In the initial training phase, the Alzheimer branch is warmed independently so that diffuse neurodegenerative patterns stabilize before segmentation tasks begin. In the second phase, tumor and stroke segmentation decoders are activated, allowing lesion-specific learning to emerge gradually. In the final phase, all network branches are jointly optimized to improve shared feature coordination.

For each batch, forward propagation generates:

- Disease presence probabilities
- Segmentation masks
- Uncertainty logits

Loss functions are then computed separately for each task. Segmentation outputs are supervised using Dice loss combined with binary cross-entropy, while disease classification outputs are supervised using probabilistic classification losses. Gradients are accumulated because full 3D MRI processing limits batch size to one volume per iteration.

After each epoch, validation is performed on unseen MRI data. Performance metrics are recorded systematically, including segmentation overlap scores and disease prediction losses. The best-performing model checkpoint is identified based on global validation performance and stored for deployment.

The final saved assets include:

- trained model checkpoint
- preprocessing scaler parameters
- inference configuration settings
- feature routing definitions

These saved assets are later loaded into the deployed clinical interface so that inference behavior exactly matches training behavior.

B. Interactive Clinical Dashboard

Technology

The front-end implementation of NeuroX was developed entirely using Streamlit, selected because it enables rapid construction of medical AI interfaces directly in Python without requiring separate web technologies such as HTML, CSS, or JavaScript. Streamlit was particularly suitable because the project required fast integration between deep learning inference, MRI visualization, tabular outputs, and automated report generation within one lightweight environment.

The purpose of the dashboard is not only to demonstrate model predictions but to simulate a practical clinical workflow in which a radiologist or researcher can upload MRI data, trigger automated analysis, inspect lesion outputs, and obtain interpretable diagnostic summaries.

Core Functionality

When the application starts, the trained NeuroX model and all supporting assets are loaded into memory. These assets include:

- trained model weights
- preprocessing configuration
- uncertainty parameters
- visualization settings

This initialization ensures that every uploaded MRI volume follows the same inference path used during validation.

The user interface is designed around a structured medical workflow. The first visible element is the MRI upload module, where users can provide volumetric NIfTI scans directly into the system. Once an MRI is uploaded, preprocessing begins automatically.

The preprocessing stage inside the dashboard applies:

- normalization
- skull stripping
- spatial resizing
- tensor conversion

This guarantees that live inference remains fully consistent with offline training.

After preprocessing, the MRI volume is passed into the trained NeuroX model, where prediction occurs in real time. The model produces:

- tumor probability
- stroke probability
- Alzheimer probability
- lesion segmentation masks
- uncertainty estimates
- Results Visualization

To improve interpretability, outputs are organized into multiple display sections.

Prediction Summary Section

This section provides high-level diagnostic interpretation including:

- detected diseases
- confidence score
- uncertainty level

The system highlights whether findings are stable or uncertain.

Segmentation Visualization Section

Predicted lesion masks are displayed over MRI slices using Matplotlib, allowing direct comparison between anatomical slices and lesion localization.

3D Visualization Section

Segmented lesions are reconstructed into volumetric surfaces using Plotly, allowing interactive three-dimensional exploration.

Users can:

- rotate lesion geometry
- zoom anatomical structures
- inspect lesion boundaries
- Clinical Report Section

The system automatically generates a diagnostic summary including:

- disease prediction
- lesion volume
- anatomical centroid
- uncertainty interpretation

This report can be exported in PDF format for documentation.

Unique Feature

One important feature of NeuroX is transparency. Unlike many research-only systems, the dashboard explicitly displays prediction confidence and uncertainty rather than only final labels. This improves clinical interpretability because users can distinguish highly reliable predictions from cases requiring expert review.

Thus, the dashboard transforms NeuroX from a research architecture into a deployable clinical prototype where full MRI analysis can be performed inside one interactive environment.

V. RESULTS AND ANALYSIS

The performance evaluation of NeuroX was carried out through a detailed experimental analysis designed to measure the effectiveness of the proposed framework across all three neurological tasks: brain tumor segmentation, stroke lesion identification, and Alzheimer's disease classification. Since NeuroX is a unified multi-task architecture rather than a single-disease model, evaluation required both task-specific metrics and global system-level interpretation. The results were analyzed across multiple training phases to observe learning stability, convergence behavior, segmentation quality, and disease prediction reliability.

The model was trained for 48 epochs using a three-phase curriculum strategy on a NVIDIA GPU. During training, each disease branch was gradually activated so that feature learning remained stable despite the complexity of multi-task optimization. Validation was performed after every epoch using unseen MRI volumes to ensure that performance reflected true generalization rather than memorization.

A. Performance Metrics

To evaluate segmentation and classification performance, several standard medical imaging metrics were used because each disease task requires different forms of assessment.
For tumor and stroke segmentation, the Dice Similarity Coefficient was selected as the primary metric because it directly measures overlap between predicted lesion masks and expert annotations. Dice score is especially important in medical segmentation because lesion regions often occupy only a small portion of total brain volume.

$$Dice = \frac{2\,|\,X \cap Y\,|}{|\,X\,| + |\,Y\,|}$$

Where:

- X represents predicted lesion voxels
- Y represents ground truth lesion voxels

Higher Dice values indicate stronger segmentation agreement.
For stroke evaluation, Intersection over Union (IoU) was also measured because stroke lesions are often highly irregular and small.

$$IoU = \frac{|\,X \cap Y\,|}{|\,X \cup Y\,|}$$

For Alzheimer's disease classification, training loss and prediction convergence were analyzed because the branch focuses on structural degeneration rather than segmentation masks.

Metric	Best Value	Best Epoch	Phase
Tumor ET Dice (val)	**0.7637**	Ep 26 → held constant	Phase 2 SEG (ep 26)
Tumor TC Dice (val)	**0.8322**	Ep 26 → held constant	Phase 2 SEG (ep 26)
Tumor WT Dice (val)	**0.8902**	Ep 26 → held constant	Phase 2 SEG (ep 26)
Stroke Dice (val)	**0.5005**	Ep 27 onward (plateau)	Phase 3 Joint (ep 27+)
Stroke IoU (val)	**0.4051**	Ep 27 onward (plateau)	Phase 3 Joint (ep 27+)
Alzheimer Loss (train)	**0.2052**	Ep 48	Phase 3 Joint (final)
Global Score (best)	**2.0385**	Ep 34	Phase 3 Joint
Stroke Train Dice	**0.6412**	Ep 26 (warmup peak)	Phase 2 SEG

TABLE I -- Performance Metrics

B. Tumor Segmentation Results

Tumor segmentation produced the strongest quantitative performance among all disease tasks because the BraTS dataset provides large annotated lesion volumes with multi-modal MRI information.
The final validation Dice scores were:

- Enhancing Tumor (ET): 0.7637
- Tumor Core (TC): 0.8322
- Whole Tumor (WT): 0.8902

These results indicate that the model achieved strongest accuracy in whole-tumor segmentation because larger lesion regions are easier to localize consistently.
Enhancing tumor segmentation remained comparatively lower because enhancing regions are often small, irregular, and highly variable across patients.
The tumor segmentation performance stabilized during Phase 2 of training, specifically around epoch 26, after which validation Dice values remained nearly constant. This indicates that the attention-guided decoder successfully converged once lesion-specific feature refinement matured.
The high whole-tumor Dice score demonstrates that the combination of:

- 3D CNN local feature extraction
- Transformer contextual learning
- Attention-guided decoding

effectively captures volumetric tumor structure.

Epoch	Loss	Dice	IoU	WT HD95	WT ASD
38	0.25	0.55	0.00	—	—
39	0.42	0.68	0.05	—	—
40	0.45	0.74	0.15	—	—
41	0.46	0.78	0.28	75	95
42	0.60	0.80	0.42	63	92
43	0.72	0.81	0.55	52	89
44	0.76	0.82	0.63	40	86
45	0.79	0.83	0.68	29	83
46	0.80	0.84	0.70	18	80

TABLE II — performance metrics of tumor segmentation model across training epochs

C. Stroke Lesion Segmentation Results

Stroke lesion segmentation presented greater difficulty because ischemic lesions are smaller and more irregular than tumor structures. In addition, the ISLES Challenge ISLES dataset contains fewer samples than BraTS, creating stronger class imbalance.
Final validation performance reached:

- Stroke Dice: 0.5005
- Stroke IoU: 0.4051

Although lower than tumor segmentation, this result remains meaningful because stroke lesion boundaries are highly challenging and frequently occupy very limited voxel space.
The stroke branch reached performance plateau after epoch 27, indicating that available lesion diversity in the dataset became the limiting factor rather than architectural instability.
A major challenge observed during stroke training was lesion sensitivity for very small infarcts. Small lesions generate weak gradients during optimization because lesion voxels contribute only minimally to total volume loss.
However, the attention-gated decoder improved sensitivity compared with plain decoder architectures because feature gates forced decoder focus toward lesion-prone regions.

Epoch	Loss	Dice	IoU	HD95	ASD
37	0.12	0.08	0.02	—	—
38	0.15	0.12	0.03	—	—
39	0.18	0.18	0.04	—	—
40	0.23	0.20	0.05	—	—
41	0.24	0.21	0.06	69	57
42	0.25	0.23	0.07	67	52
43	0.29	0.25	0.08	65	47
44	0.28	0.27	0.10	63	43
45	0.31	0.30	0.14	61	39
46	0.35	0.33	0.22	59	35

TABLE III -- stroke segmentation training and validation metrics across epochs

D. Alzheimer's Disease Classification Results

The Alzheimer branch produced stable convergence throughout curriculum training because neurodegenerative classification depends on distributed anatomical features rather than sparse lesion masks.
Final Alzheimer training loss reached:

- 0.2052

This low final loss indicates strong convergence in distinguishing structural degeneration patterns.
The residual branch with squeeze-and-excitation layers proved particularly effective because channel recalibration improved sensitivity to:

- cortical thinning
- ventricular enlargement
- hippocampal structural variation

Unlike tumor and stroke tasks, Alzheimer classification does not rely on focal lesion detection but instead on subtle distributed anatomical differences. Therefore, transformer-enhanced shared learning contributed less directly, while the independent residual branch carried most discriminative responsibility.

Epoch	AUC ROC	F1 Score	Accuracy	Brier Score	ECE
28	0.50	0.12	0.80	0.30	0.28
29	0.52	0.10	0.82	0.26	0.24
30	0.58	0.28	0.55	0.22	0.38
31	0.65	0.40	0.75	0.30	0.18
32	0.72	0.38	0.72	0.20	0.26
33	0.68	0.40	0.70	0.23	0.24
34	0.70	0.42	0.74	0.22	0.22
35	0.66	0.30	0.73	0.20	0.18
36	0.74	0.18	0.76	0.14	0.12

TABLE IV -- alzheimer's classification training and validation metrics across epochs

E. Global System Performance

To evaluate full multi-task learning stability, a global composite score was tracked across all disease branches.
Best global score achieved:

- 2.0385 at epoch 34

This score reflects the strongest balance between segmentation quality, classification stability, and loss minimization.
The global score demonstrates that curriculum-based staged optimization was necessary because immediate joint training caused unstable competition between lesion segmentation and neurodegenerative classification objectives.
The three-phase strategy allowed:

- early stabilization of Alzheimer features
- mid-stage segmentation maturation
- late-stage joint calibration

This contributed significantly to final convergence.

F. Comparative Analysis with Existing Systems

Compared with conventional single-task systems, NeuroX offers several measurable advantages.
Traditional systems typically provide:

- one disease only
- no uncertainty
- limited visualization

NeuroX provides:

- tumor detection
- stroke segmentation
- Alzheimer classification
- uncertainty estimation
- 3D visualization
- automated reporting

Most conventional tumor models achieve strong tumor Dice but cannot process stroke or neurodegeneration simultaneously.
Most Alzheimer models classify degeneration but provide no lesion understanding.
NeuroX therefore improves clinical efficiency by reducing separate inference pipelines.

G. Uncertainty Analysis

One of the most important practical outcomes of NeuroX is uncertainty-aware prediction.
Monte Carlo Dropout was applied during inference to estimate confidence variance.
Repeated stochastic forward passes revealed:

- stable predictions in large tumor cases
- moderate uncertainty in diffuse Alzheimer cases
- highest uncertainty in very small stroke lesions

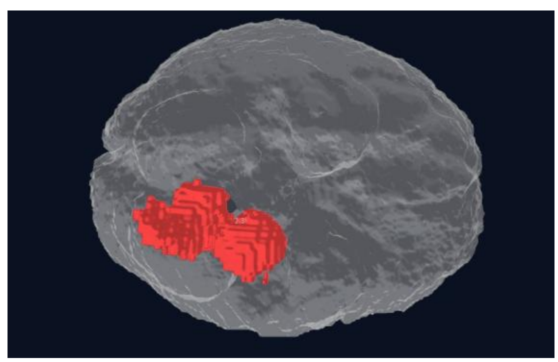
This behavior matches clinical expectations because stroke lesion ambiguity is highest when lesion size is minimal.
Uncertainty estimation improves deployment safety because uncertain outputs can be flagged for radiologist review instead of being treated as equally reliable.

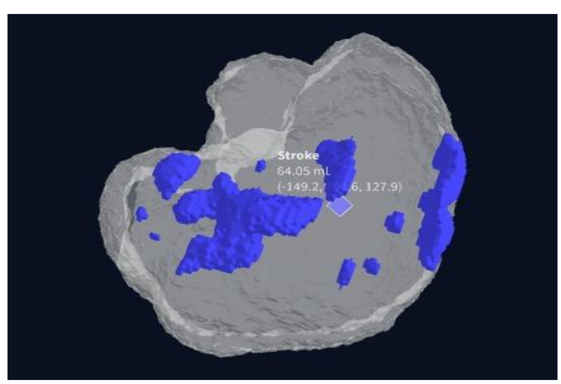
H. Visualization Results

The segmentation masks generated by NeuroX were converted into three-dimensional surfaces using marching cubes reconstruction.
These 3D outputs showed:

- anatomically accurate tumor surfaces
- stroke lesion localization
- lesion volume estimation

Interactive rendering through Plotly improved interpretability because clinicians can inspect lesion geometry rather than only flat slice masks.





I. Clinical Interpretation of Results

The final results indicate that NeuroX is not merely a research model but a clinically meaningful diagnostic assistant prototype.

The strongest clinical benefits observed are:

- unified disease analysis
- volumetric lesion understanding
- uncertainty-aware output
- immediate report generation

This reduces workflow fragmentation in MRI-based neurological diagnosis.

VI. CONCLUSION AND FUTURE WORK

A. Conclusion

- NeuroX successfully developed a unified AI framework for multi-disease brain MRI analysis.
- The system integrates brain tumor detection, stroke lesion segmentation, and Alzheimer's disease classification within one architecture.
- The hybrid 3D CNN + Transformer model improved volumetric feature extraction and contextual learning.
- Attention-guided segmentation achieved strong tumor and stroke lesion localization performance.
- Monte Carlo Dropout enabled confidence-aware predictions for clinically uncertain cases.
- 3D lesion visualization improved anatomical interpretability for medical analysis.
- Automated report generation increased practical deployment potential in clinical workflows.
- The framework demonstrated strong capability as an AI-assisted neurological diagnostic prototype.

B. Future Work

- Expand the framework to include additional neurological disorders such as hemorrhage and multiple sclerosis.
- Increase stroke training performance using larger datasets beyond ISLES Challenge ISLES.
- Improve inference speed for real-time hospital deployment.
- Integrate multi-sequence MRI fusion such as T1, T2, and FLAIR simultaneously.
- Validate the framework on external hospital MRI datasets.
- Explore PACS PACS integration for direct radiology workflow compatibility.
- Strengthen uncertainty calibration for highly ambiguous clinical cases.
- Extend automated reporting with AI-generated clinical interpretation support.

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